Image Transforms-Spectral

Multi-band Operations

Arithmetic Operations

Image Transform

- **Image Transform**
  - **An operation that re-expresses the information content of an image or an image set**
- **Spectral Transform**

  Use of spectral information of the object recorded in multiple bands

Multi-band Operations

- **Allow the generation of a derived image from two or more bands of a multispectral or multi-temporal image**
- **To enhance or extract features from satellite images which cannot be clearly detected in a single band**
- **The derived image may well have properties that make it more suited to a particular purpose than the original**

Multi-band operations

- Images may be separate spectral bands from a single multi spectral data set, or they may be individual bands from data sets that have been recorded at different dates or using different sensors, BUT
  - preferably from same sensor
- The operations of addition, subtraction, multiplication and division, are performed on two or more co-registered images of the same geographical area

Arithmetic Operations

- **Addition of images**

$$G_{sum}(i, j) = [G1(i, j) + G2(i, j)]/2$$

  - **Really a form of averaging**
  - **Averaging can be carried out on multiple images of the same area in order to reduce the noise component**

Arithmetic Operations

- **Subtraction**
  - $G_{absdiff}(i, j) = |G1(i, j) - G2(i, j)|$

- **Is used to reveal differences between those images**
- **Is often used in the detection of change if the images involved were taken at different dates**
- **Performed on pixel by pixel basis**
- **The vertical bars | · | denote the absolute value (regardless of sign)**

Arithmetic Operations

- **Multiplication**
- **Pixel-by-pixel multiplication of two images - rarely in practice**
- **Normally involves the use of a single ‘real’ image and a binary image made up of ones and zeros**
- **The binary image is used as a mask, for those image pixels in the real image that are multiplied by zero also become zero, while those that are multiplied by one remain the same**
- **Useful if an image of interest is composed of two or more distinctive regions and if the analyst is interested only in one of these regions**

Arithmetic Operations

- **Division or ratioing of images**
  - **Is probably the arithmetic operation that is most widely applied to images in various applications of remote sensing**
  - **Division operation is used to detect the magnitude of the differences between spectral bands**
  - **Ratio images are enhancements resulting from the division of DN values in one spectral band by the corresponding values in another band**

Band Ratio

$$BV_{i,j,ratio} = \frac{BV_{i,j,k}}{BV_{i,j,l}}$$

- **where:**
  - $BV_{i,j,k}$ **is the original input brightness value in band** ***k***
  - $BV_{i,j,l}$ **is the original input brightness value in band** ***l***
  - $BV_{i,j,ratio}$ **is the ratio output brightness value**

Spectral Ratioing

- **A major advantage of ratio images is that they convey the spectral or color characteristics of image features, regardless of variations in scene illumination conditions/ topography**

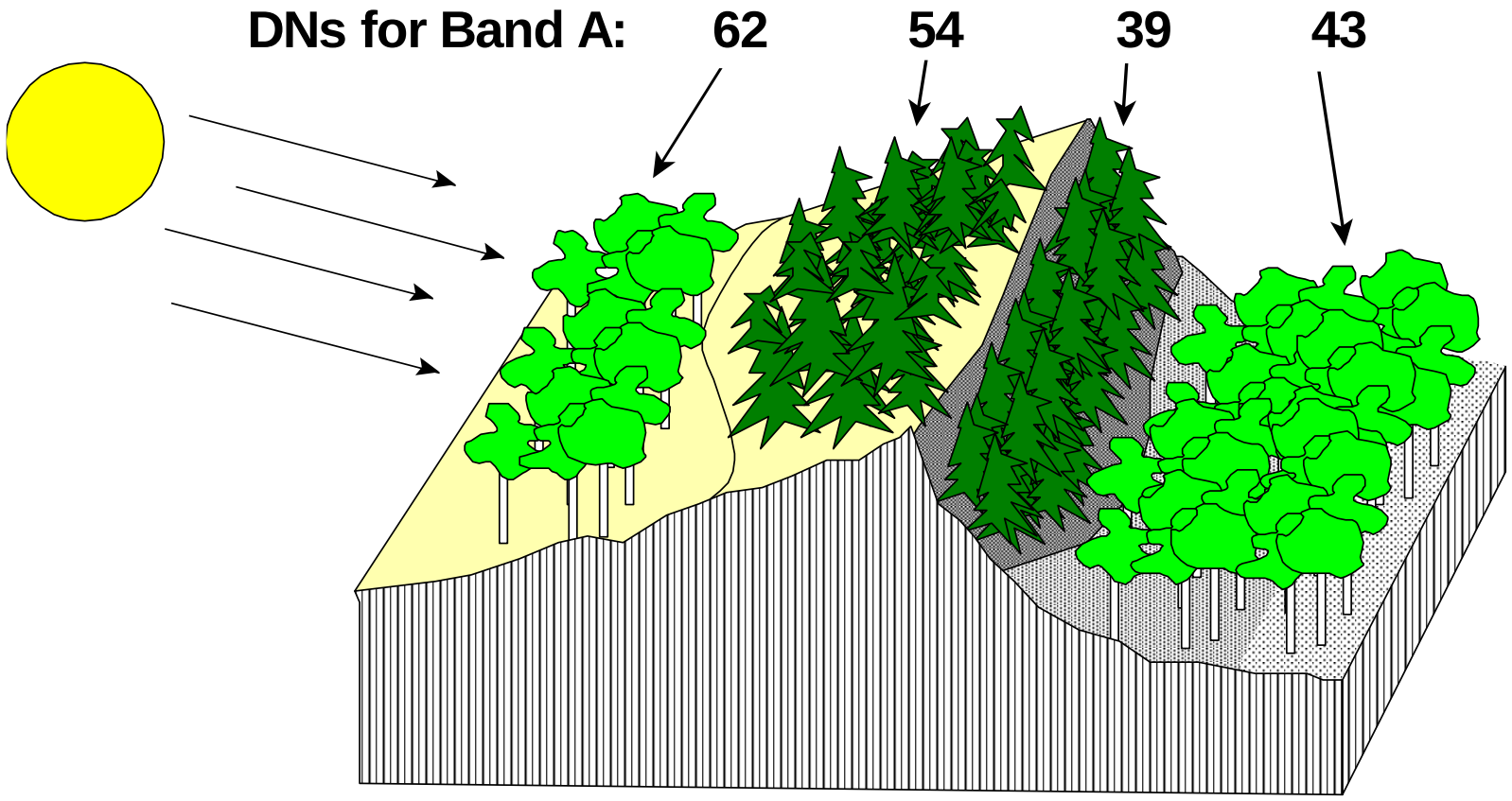
Spectral Ratioing

- **Ratios are effective in enhancing or revealing the information when there is an inverse relationship between two spectral responses to the same biophysical phenomenon**
- **A major advantage of ratio images is that they convey the spectral or color characteristics of image features, regardless of variations in scene illumination conditions / topography**

Image ratios: Brightness variations

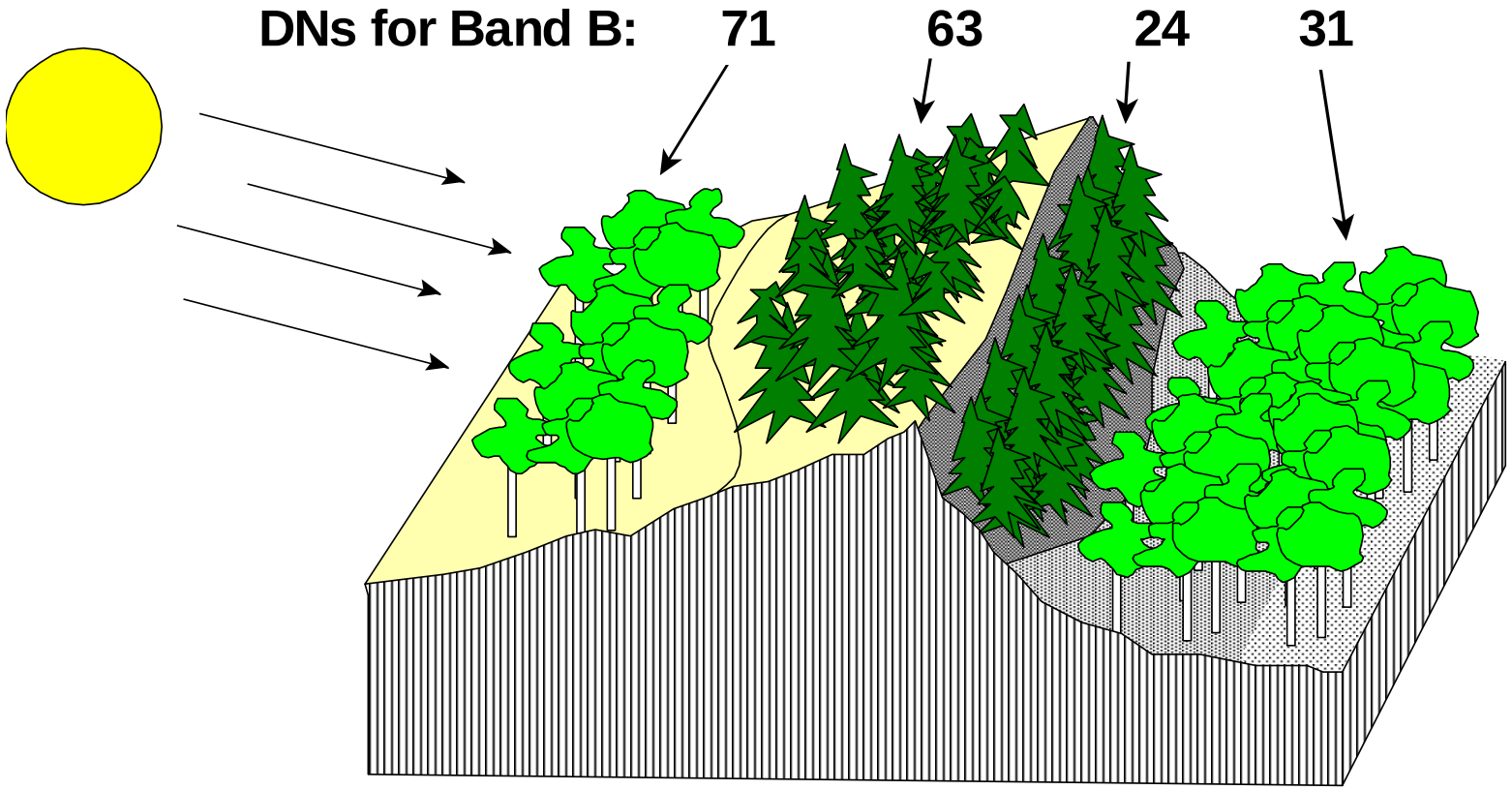
- **When a satellite passes over an area with relief, it records both shaded and sunlit areas**
- **The observed DNs in the individual bands are substantially lower on the shaded side compared to the sunlit side**
- **However, the ratio values are nearly identical, irrespective of illumination conditions**
- **A ratio image of the scene effectively compensates for the brightness variation, caused by the differences in the topography**

Varying Illumination

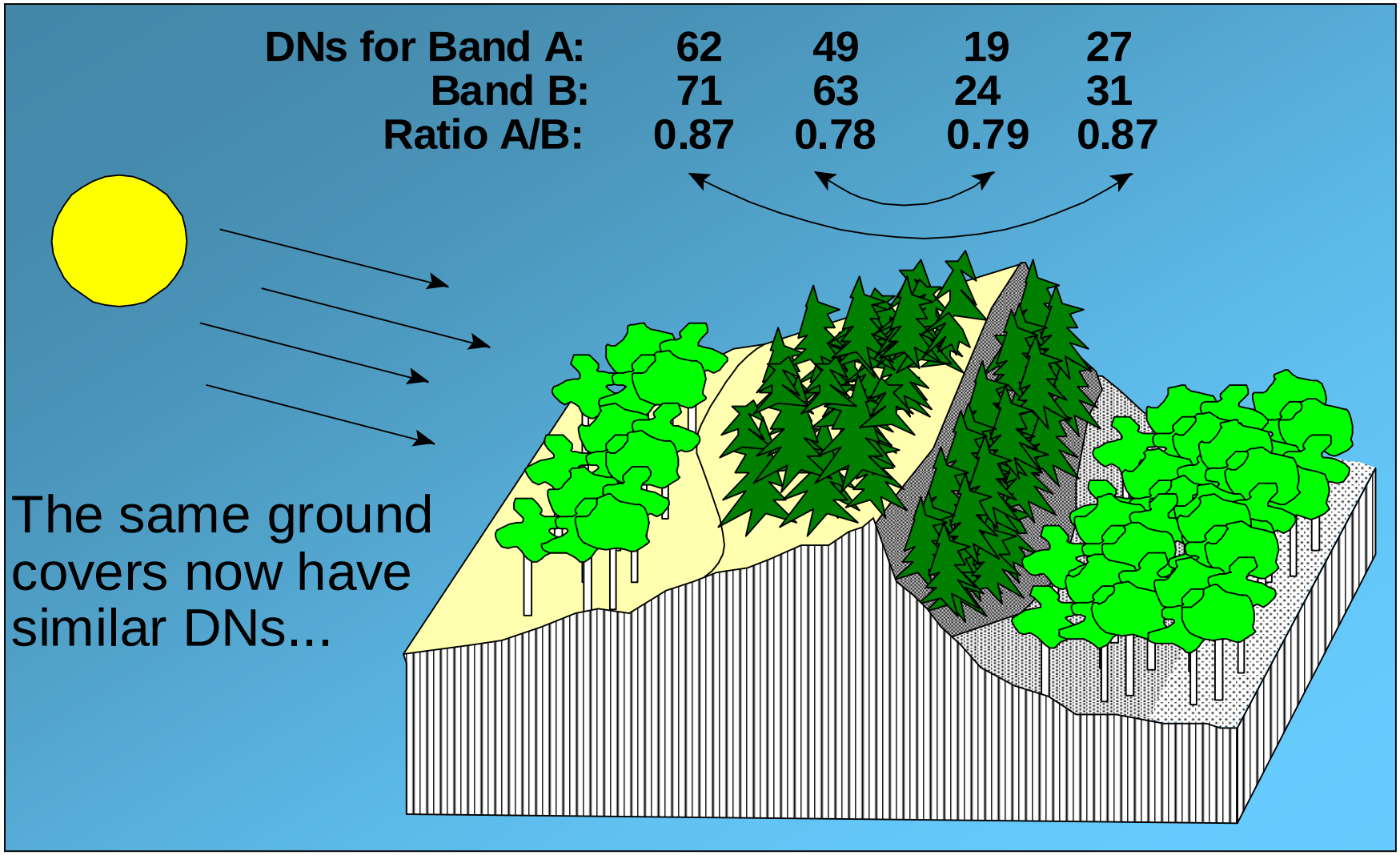


The same ground cover has different DNs...

Band B has the same “Problem”



Varying Illumination & Spectral Ratios



Scene Illumination Conditions

Benefit of spectral ratioing

- Its looking at relative values (i.e.ratios) instead of absolute brightness values
- Variations in scene illumination as a result of topographic effects are reduced
- Although the absolute reflectance for forest covered slopes may vary depending on their orientation relative to the sun's illumination, the ratio of their reflectance between the two bands should always be very similar

Band Ratio

- **It is also important to note that the ratio between two raw DNs for a pixel will normally be quite different from that between two radiance values computed for the same pixel**
- **Some trial and error may also be necessary for this particular selection to obtain best output**

Ratio Composites

- Ratio images can also be used to generate false color composites by combining three monochromatic ratio datasets
- Such composites have advantage of combining data from more than two bands and presenting the data in color which further facilitates the interpretation of slight spectral reflectance differences
- Selection of ratios to include in a composite and selecting colors in which to portray them can sometimes be difficult
- Some trial and error is often necessary in selecting ratio combinations

Reference/Reading Assignment

- **Computer Processing of Remotely sensed Images, Mather,**
  - **Chapter 6, sec 6.2**

